



EyeArt artificial intelligence analysis of diabetic retinopathy in retinal screening events

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Abstract

Purpose Early detection and treatment of diabetic retinopathy (DR) are critical for decreasing the risk of vision loss and preventing blindness. Community vision screenings may play an important role, especially in communities at higher risk for diabetes. To address the need for increased DR detection and referrals, we evaluated the use of artificial intelligence (AI) for screening DR.

Methods Patient images of 124 eyes were obtained using a 45° Canon Non-Mydriatic CR-2 Plus AF retinal camera in the Department of Endocrinology Clinic (Newark, NJ) and in a community screening event (Newark, NJ). Images were initially classified by an onsite grader and uploaded for analysis by EyeArt, a cloud-based AI software developed by Eyenuk (California, USA). The images were also graded by an off-site retina specialist. Using Fleiss kappa analysis, a correlation was investigated between the three grading systems, the AI, onsite grader, and a US board-certified retina specialist, for a diagnosis of DR and referral pattern.

Results The EyeArt results, onsite grader, and the retina specialist had a 79% overall agreement on the

diagnosis of DR: 86 eyes with full agreement, 37 eyes with agreement between two graders, 1 eye with full disagreement. The kappa value for concordance on a diagnosis was 0.69 (95% CI 0.61–0.77), indicating substantial agreement. Referral patterns by EyeArt, the onsite grader, and the ophthalmologist had an 85% overall agreement: 96 eyes with full agreement, 28 eyes with disagreement. The kappa value for concordance on “whether to refer” was 0.70 (95% CI 0.60–0.80), indicating substantial agreement. Using the board-certified retina specialist as the gold standard, EyeArt had an 81% accuracy (101/124 eyes) for diagnosis and 83% accuracy (103/124 eyes) in referrals. For referrals, the sensitivity of EyeArt was 74%, specificity was 87%, positive predictive value was 72%, and negative predictive value was 88%.

Conclusions This retrospective cross-sectional analysis offers insights into use of AI in diabetic screenings and the significant role it will play in automated detection of DR. The EyeArt readings were beneficial with some limitations in a community screening environment. These limitations included a decreased accuracy in the presence of cataracts and the functional cost of EyeArt uploads in a community setting.

Keywords Artificial intelligence · Community vision outreach · Diabetic retinopathy

Abbreviations

AI Artificial intelligence
BMI Body mass index

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DLAI	Deep learning artificial intelligence
DR	Diabetic retinopathy
T2DM	Type II diabetes mellitus
VTDs	Vision-threatening diseases

Introduction

In 2019, 37.3 million Americans, had diabetes, of whom 90–95% had Type II Diabetes Mellitus (T2DM) [1]. An additional estimated 96 million individuals aged 18 and older in the USA were prediabetic (for T2DM) [2], and there are 1.4 million new cases of T2DM diagnosed annually [1]. However, surveillance studies have estimated that 23% of adults with T2DM remain undiagnosed [3]. T2DM prevention, screening, and management has been at the forefront public health efforts both in the USA and worldwide [4, 5], as it is associated with many health complications, including heart disease, kidney disorders, depression, lack of mobility, and strokes. Individuals with T2DM are also at elevated risk for developing diabetic retinopathy (DR). Hyperglycemia-related retinal vascular injury results in retinal microaneurysms, retinal hemorrhages, edema, vitreous hemorrhage, and tractional retinal detachment, all of which can cause vision loss and eventually blindness. Almost 9% of adults over age 45 with diagnosed diabetes have DR, and 4% experience vision loss from DR [6]. Individuals with diabetes have a higher incidence of other vision-threatening diseases (VTDs): cataracts are noted in 32.2%, glaucoma in 7.1%, and age-related macular degeneration in 4.3% [6].

Patient education regarding ocular complications of DM can play a significant role in the early diagnosis and treatment of DR. Despite the significant impact, DR can have on quality of life, diabetic eye examination compliance is poor, with estimates ranging from 18 to 60% [7]. DR screening programs, typically involving fundus photography and review by a specially—telemedicine grader, have become prevalent to address increasing rates of DR and maintain adequate community surveillance [8]. Disparities in access to diabetes management and DR screenings still persist, as racial minorities tend to present with more advanced stages of DR from a lack of

appropriate screening [9]. Other risk factors include lack of health literacy and gender/age disparities [10].

Health-care providers have turned toward new solutions to address these deficits in DR screening in management. Deep Learning Artificial Intelligence (DLAI) is a type of artificial intelligence technology that utilizes multiple layers of analysis to abstract information from images in a data set. DLAI has been deployed throughout health care to increase the efficiency and reach of available services. For instance, IDx-DR, (Digital Diagnostics, Iowa, USA), became FDA-approved in April of 2018 for the autonomous detection of DR with imaging using a Topcon NW400. IDx was designed to detect the presence of absence of DR [11]. More recently in 2020, EyeArt (EyeNuk, California, USA), became the first DLAI system FDA-approved for the detection of both more than mild and vision-threatening DR. EyeArt is compatible with the Canon CR-2 Plus AF digital non-mydratic retinal camera (Tokyo, Japan) and uses uploaded retinal images to classify the images as: DR not present, more than mild DR, or vision-threatening DR [12]. Previous EyeArt validation studies have also demonstrated use of the software by novice operators, therefore making EyeArt suitable for increased community screening outreach [12]. The utilization of this technology to detect DR has a reported sensitivity of 91.3% and a specificity of 91.1% [13].

Early detection, intervention, and education in some cases can limit or reverse the progression of DR. Therefore, evaluation of new technologies to improve the diagnosis process is critical. In times such as COVID-19 pandemic, when regular in-person visits may be difficult to achieve, implementation of an AI system that reduces close contact between individuals may help to address gaps in access to eye care. The objective of this study was to evaluate an artificial intelligence system, EyeArt, as a DR screening tool in an underserved urban community. We determined sensitivity and specificity of EyeArt using the referrals of a board-certified ophthalmologist as the gold standard. We also assessed the concordance of EyeArt with both a telemedicine grader and a board-certified retina specialist for both the presence of DR and the need for referral.

Methods

We retrospectively evaluated the fundus images from community diabetic retinal screenings conducted in 2022: one screening from an endocrine clinic with diabetics and one from a screening conducted in an urban community center. A total of 62 patients were identified who all underwent non-mydriatic retinal imaging screening using a Canon CR-2 Plus AF Retinal Camera with a resolution of 23 mega-pixel (Tokyo, Japan). To evaluate AI in identifying DR, criteria for patient inclusion included age over 18 years old and a record of at least two posterior pole retinal images (one macula-centered, one optic nerve-center) and one anterior segment image per eye. The cost to cover screening equipment, AI, and van to transport equipment had been obtained through grant funding.

Each patient record from the retinal camera consisted of three images per eye. The first was an

anterior segment image, and the second and third were posterior pole (retinal) images: one macula-centered, and one optic nerve-centered. EyeArt requires four retinal images (macula-centered and optic nerve-centered for each eye) for single patient analysis. Patient information was entered into EyeArt, and diabetes status (Type 2 Diabetes or Unknown) was selected. The retinal images for 124 eyes were uploaded to EyeArt, which produces a report that characterizes each eye as “more than mild DR”, “vision-threatening DR”, negative for both more than mild and vision-threatening DR, or “ungradable” (Fig. 1). If EyeArt detects an image of “poor quality” or “insufficient retinal coverage”, it produces the “ungradable” output⁶. The human graders were blinded to the results produced by the AI at the time of imaging grading. The onsite grader evaluated posterior pole images for DR grading and used the WHO criteria to grade the anterior segment images for



EyeArt
 Diabetic Retinopathy Analysis Report

Patient Information		General Information	
Patient ID:		Referring Location:	
Patient Name:		Referring Provider:	
Date of Birth:		EyeArt Control ID:	
Gender:		Dilation Status:	
Encounter ID:		Exam Analysis Date:	

EyeArt Diabetic Retinopathy (DR) Exam Result Summary
 Negative for more than mild DR in both eyes. Return for retinal imaging within 12 months.

Right Eye Results	Left Eye Results
mtmDR Result: Negative for more than mild DR	mtmDR Result: Negative for more than mild DR
vtDR Result: Negative for vision-threatening DR	vtDR Result: Negative for vision-threatening DR




Macula Centered, Right Eye Macula Centered, Left Eye




ONH Centered, Right Eye ONH Centered, Left Eye

*Do not use the above thumbnail images for diagnostic purposes.

Notes
 A negative result indicates a low risk for moderate non-proliferative DR, severe non-proliferative DR, proliferative DR, and diabetic macular edema.

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EyeArt
 Diabetic Retinopathy Analysis Report

Patient Information		General Information	
Patient ID:		Referring Location:	
Patient Name:		Referring Provider:	
Date of Birth:		EyeArt Control ID:	
Gender:		Dilation Status:	
Encounter ID:		Exam Analysis Date:	

EyeArt Diabetic Retinopathy (DR) Exam Result Summary
 Vision-threatening DR detected in both eyes. Refer to an eye care professional for evaluation (with preferential scheduling if possible).

Right Eye Results	Left Eye Results
mtmDR Result: More than mild DR detected	mtmDR Result: More than mild DR detected
vtDR Result: Vision-threatening DR detected	vtDR Result: Vision-threatening DR detected




Macula Centered, Right Eye Macula Centered, Left Eye




ONH Centered, Right Eye ONH Centered, Left Eye

*Do not use the above thumbnail images for diagnostic purposes.

Notes
 A positive result for vision-threatening diabetic retinopathy indicates a high risk for severe non-proliferative DR, proliferative DR, or diabetic macular edema.

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Fig. 1 Sample reports provided by EyeArt (California, USA) for healthy eyes versus eyes with DR

cataracts [14]. Posterior pole images were later evaluated by a board-certified retina specialist and marked the eyes for referral for clinical retinal examination. Diagnoses by the onsite grader and retina specialist were either “positive for DR,” “negative for DR,” or “ungradable.” Each patient’s record had three reports that were analyzed and compared—from AI by EyeArt, the onsite grader, and the off-site retinal specialist.

Concordance between EyeArt, the onsite grader, and retina specialist was evaluated for both the DR diagnosis and a referral for a clinical retinal examination was made (Table 1). Diagnosis was considered concordant when the human graders designated “positive for DR” and EyeArt identified any severity level of DR (more than mild or vision-threatening). Classification of “negative for DR” by the human graders was considered the same as “negative for more than mild DR, or, negative for vision-threatening DR” by EyeArt. Images marked ungradable by human graders and EyeArt were also considered to be concordant. Referral for a clinical diabetic retinal examination by human graders was categorized as “referral” or “no referral.” EyeArt’s classification of any severity level of DR or ungradable was considered concordant with “referral” by the human graders. EyeArt classification of “negative for more than mild DR, negative for vision-threatening DR” agreed with the “no referral” by the human graders.

The results produced by EyeArt were compared to the reports from the onsite grader at the screening events and a retina specialist. Free-marginal multi-rater kappa values and the associated confidence intervals were calculated using free-marginal Fleiss kappa analysis to evaluate the agreement between the AI, onsite grader, and ophthalmologist (Table 2). The accuracy, sensitivity, specificity, positive predictive

Table 2 Interpretation of kappa values

Kappa value	Interpretation of kappa value
≤ 0	No agreement
0.1–0.20	Slight agreement
0.21–0.40	Fair agreement
0.41–0.60	Moderate agreement
0.61–0.80	Substantial agreement
0.81–0.99	Near perfect agreement
1	Perfect agreement

value, and negative predictive value of EyeArt referrals were also calculated compared to the ophthalmologist. Analysis was conducted for diagnosis rates and for referral rates separately. This study was approved by the Rutgers University Institutional Review Board (Pro20140001070) and is in compliance with the World Medical Association’s Declaration of Helsinki. All participants provided verbal informed consent.

Results

Our tele-ophthalmology community screenings over the last six months assessed 124 eyes of 62 patients with a mean age of 64 years old. Thirty-eight eyes were examined from the endocrine clinic screening, and 86 eyes were assessed from the community center screening. Forty-five percent of patients had confirmed or self-reported T2DM, 32% reported no history of T2DM, and 23% of patients did not provide this information. Twenty-seven (44%) patients were males, and 34 were females, with an average self-reported BMI of 29.7. Sixty-nine percent of the patients were Caucasian (Portuguese), 21% Hispanic, and 10% were African American (Table 3).

Table 1 Classification of concordance between EyeArt and human graders (retinal specialist and onsite grader)

	Human grader	EyeArt
Diagnosis	“Positive for DR”	“More than mild DR” OR “Vision-threatening DR”
	“Negative for DR”	“Negative for more than mild DR, negative for vision-threatening DR”
	“Ungradable”	“Ungradable”
Referrals	“Referral”	“More than mild DR” OR “Vision-threatening DR” OR “Ungradable”
	“No referral”	“Negative for more than mild DR, negative for vision-threatening DR”

DR diabetic retinopathy

Table 3 Demographic characteristics of patients evaluated for diabetic retinopathy

Average BMI	29.7 (25–29.9: overweight)
Average male BMI	29.5 (25–29.9: overweight)
Average female BMI	29.9 (25–29.9: overweight)
Race/ethnicity	Proportion of patients
Caucasian (Portuguese)	0.69
Hispanic	0.21
African American	0.10

BMI body mass index

Table 4 Proportion of patients with positive DR diagnosis by EyeArt, onsite grader, and board-certified ophthalmologist (retina specialist)

	Patients with a positive DR diagnosis <i>N</i> (Proportion)	Patients with an ungradable image <i>N</i> (Proportion)
EyeArt	15 (0.24)	7 (0.11)
Onsite grader	17 (0.27)	1 (0.02)
Board-certified ophthalmologist	15 (0.24)	9 (0.15)

DR diabetic retinopathy

Retinal fundus photographs images were assessed for DR by an onsite grader, retina specialist, and by AI, EyeArt. Fundus photos uploaded to EyeArt yielded positive DR in 15 (24%) patients. The onsite grader found DR in 17 (27%) patients, and the retina specialist found DR in 30 (24%) patients (Table 4). Seven patients (11%) had at least one image that was ungradable by EyeArt, compared to only one (2%) by the onsite grader and nine (15%) by the retina specialist (Table 4). Excluding patients the grader found

ungradable images for both eyes, AI found DR in at least one eye in 12 (50%) patients with confirmed or self-reported T2DM, none (0%) in the patients who denied T2DM, and three (25%) of patients with unknown status. Similarly, the onsite grader found DR in at least one eye in 12 (43%) patients with confirmed or self-reported T2DM, two (13%) patients who denied T2DM, and three (17%) of patients with unknown status. The retina specialist found DR in at least one eye in 12 (55%) patients with confirmed or self-reported T2DM and none (0%) in the patients who denied T2DM or had unknown status (Table 5).

The EyeArt results, onsite grader, and ophthalmologist produced a 79% overall agreement on diagnosis of the eyes (Fig. 2): 86 eyes with full agreement, 37 eyes with agreement between two graders, and 1 eye with full disagreement. This included results where the AI and ophthalmologist agreed on a diagnosis (positive or negative for DR), or both determined the images to be ungradable. The kappa value for concordance on a diagnosis was 0.69 (95% CI 0.61–0.77), indicating substantial agreement (Table 6). Similarly, referrals made by EyeArt, the onsite grader, and the ophthalmologist had an 85% overall agreement: 96 eyes with full agreement, 28 eyes with disagreement. Referral consistency defined both positive and ungradable results as referable, while negative results were not referable. The kappa value for concordance on whether to refer was 0.70 (95% CI 0.60–0.80), indicating substantial agreement (Table 6).

When compared to the gold standard clinical examination by the board-certified retina specialist, EyeArt had an 81% accuracy for diagnosing DR (101/124 eyes) and 83% accuracy in referrals (103/124 eyes). For referrals, the sensitivity

Table 5 Patients with a positive DR diagnosis with either confirmed or self-reported T2DM, denial of T2DM, or unknown T2DM status, excluding those that the grader deemed both eyes to be ungradable

	Patients with a positive DR diagnosis with confirmed or self-reported T2DM <i>N</i> (Proportion)	Patients with a positive DR diagnosis who denied T2DM <i>N</i> (Proportion)	Patients with a positive DR diagnosis with unknown status <i>N</i> (Proportion)
EyeArt	12 (0.50)	0 (0)	3 (0.25)
Onsite grader	12 (0.43)	2 (0.13)	3 (0.17)
Board-certified ophthalmologist	12 (0.55)	0 (0)	0 (0)

DR diabetic retinopathy, *T2DM* Type II diabetes mellitus



Fig. 2 Sample image where EyeArt marked an eye referable for DR, while both human graders would not refer it for DR

Table 6 Measure of interrater agreement between EyeArt, the onsite grader, and board-certified ophthalmologist

	Percent overall agreement (%)	Fixed marginal kappa value
Diagnosis	79.30	0.69 (95% CI 0.61–0.77)
Referral	84.95	0.70 (95% CI 0.60–0.80)

Table 7 EyeArt grading of eyes with cataracts graded 2+ or higher

	Eyes with a cataract graded 2+ or higher <i>N</i> (Proportion)	Eyes without a cataract graded 2+ or higher <i>N</i> (Proportion)
Eyes graded incorrectly or marked ungradable	18 (0.61)	106 (0.23)

of EyeArt was 74%, specificity was 87%, positive predictive value was 72%, and negative predictive value was 88%. Fifteen percent (19) of the eyes had grade 2+ or greater cataracts. EyeArt's ability to grade DR decreased by 38% in the presence of cataracts 2+ or greater (Table 7).

Discussion

Review of EyeArt in community-based retinal screenings

EyeArt AI is an analysis tool intended to improve the reach of DR screenings both within medical clinics and in community screenings. We observed a high concordance of referral rates between AI, the onsite grader, and the retina specialist, suggesting that EyeArt may be well suited for clinical use for diabetic retinal screenings. In addition, the high agreement of diagnosis suggests that EyeArt was not overly conservative in the categorization of images.

However, EyeArt's ability to grade images was not consistent across all cases. EyeArt's ability to produce a result or grade accurately decreased by 38% when the patient had a cataract graded as 2+ or greater. Older individuals, who are also more likely to be afflicted with VTDs, are more likely to be afflicted with cataracts as well. Obtaining four gradable images for the software can be a challenge for screeners. The inability to grade images in patients with cataracts is problematic since DR is more prevalent with longer duration of diabetes and may be inflicted with more severe pathology.

Eyes that were considered ungradable, many of which had cataracts, were referred to the general eye clinic. Utilization of the Dual Align™ feature, a proprietary software of the Canon retinal camera, may allow for an image with higher quality to be extracted from less informative images. After image grading, images that were previously deemed ungradable were reuploaded to AI using Opacity Suppression™, a proprietary software of the Canon retina camera. The Opacity Suppression software was useful in increasing the readability of some ungradable images. However, the need to use Opacity Suppression™ or the Dual Align™ feature presents limitations for the effectiveness of EyeArt in a community screening setting. EyeArt requires four retinal images (macula-centered and optic nerve-centered for each eye) for single patient analysis at a cost of \$23 per patient. Reimbursement of the AI use via Medicare varies between \$40 and 65 based on the state. After uploading to the cloud base software, the images cannot be altered or reuploaded. Therefore, if screeners may need to reupload images for ungradable patients after running Opacity Suppression or Dual Align,

this decreases the cost-effectiveness of the software for community-based screenings. Community-based screenings are critical in the detection of undiagnosed DR in the community. Given extenuating circumstances, such as the COVID-19 pandemic, where patient access to eye care may be further limited, accurate and cost-effective screenings are critical.

Study strengths and limitations

A major strength of our study is evaluation of EyeArt in a community that is ideal for AI DR detection. As community-based screenings are intended for patients who generally lack access to health care, we evaluated EyeArt in a population that could greatly benefit from more accessible resources. However, given the retrospective nature of our study, certain unavoidable limitations in our methodology were present. Our study focused on a population of individuals seeking eye care in a medically underserved urban community. As our study consisted of Caucasian (Portuguese), Hispanic, and African American identifying individuals with an average BMI considered to be overweight, this may limit the generalizability of our findings to the overall American population. The nature of community outreach screenings also limits the sample size and medical information available on our patients. Not all individuals who participate in screenings voluntarily share personal health information with the onsite screeners. Specific information on the duration of diabetes, medication use and adherence, and a comprehensive medical history may not be completely available. Access to additional screening tools, such as measuring intra-ocular pressures, also requires funding that may not be available. Self-selection bias of screening participants may also contribute to the trends in pathology and rates of DR observed in our study, somewhat limiting the generalizability of the results.

Review of past EyeArt deployment

We observed a 74% sensitivity and 87% specificity in referrals, as compared to a 91.3% sensitivity and 91.1% specificity in DR detection [13]. A higher specificity in referrals, along with a higher negative predictive value, indicates a greater ability to determine true negatives. This may be advantageous as compared to software with high specificity, where

there are few false positives. A false positive is preferred than a false negative in a screening setting to err on the side of caution for possible undetected DR. Discrepancies between the AI and human grader were cases that had mild DR. For some cases where the human grader classified the images as “mild DR” and not referable, the AI marked them as “more than mild” and referable (Fig. 2).

Previous evaluators of EyeArt have suggested incorporation of symptomology for the basis of referrals [15]. In addition, because EyeArt only screens for DR, patients may not be referred for ophthalmic evaluation for other co-existing VTDs. EyeArt deployment in the community therefore should not replace general ophthalmic examination for patients who may be experiencing vision deficits or be at high risk for comorbidities. EyeArt has also been deployed in primary care facilities to increase screening compliance [16]. This also can allow for more effective triage in settings where there is less access to ophthalmic specialists [16]. Providing retinal screenings in clinical settings may also allow for healthcare professionals to assist patients in counseling, finding specialists, and scheduling appointments to increase follow-up compliance. In non-clinical settings, such as community-based screenings, other telemedicine tools can be deployed to increase follow-up. For instance, tele-robotic consultation has been previously used to connect individuals with a remote physician directly after screening [17]. This lessens the burden on patients who may have lower health literacy or limited availability to resources. Developments in telemedicine within ophthalmology have been spurred by the COVID-19 pandemic, especially in the advancement of imaging modalities, incorporation of existing medical record, and improvement of communications [18]. EyeArt deployment alongside other telemedicine tools could dramatically improve the quality and scope of remote eye care delivery.

Overall incorporation of artificial intelligence in medicine has significant ethical, legal, and social implications that should be considered by healthcare practitioners. As EyeArt utilizes a cloud-based server for image analysis, ensuring HIPAA security of patient images and information is of utmost importance. Regulatory oversight of artificial intelligence is critical to the maintenance of patient privacy and informed consent. More broadly, artificial intelligence programs need to be designed using carefully curated

and inclusive data to reduce intrinsic biases and discrimination in the software. Therefore, future evaluation of EyeArt could delve into the patient perspective of artificial intelligence incorporation into their medical care. Other investigations could more specifically focus on these medico-ethical, legal, and social impacts of EyeArt in underserved communities.

Conclusion

This retrospective study found EyeArt AI to be an effective DR screening tool in our patient pool. For referrals, we observed a 0.70 kappa value of concordance (85% agreement), a sensitivity of 74%, and a specificity of 87%. For diagnoses, EyeArt, the onsite grader, and the ophthalmologist had a 0.69 kappa value of concordance (79% overall agreement). As EyeArt has more false positives, this allows for a more conservative screening approach to avoid under-referral of individuals with DR. The incorporation of AI analysis with gold standard retinal photography can expand the reach of DR awareness and care. In addition to accuracy, the speed of the system (under one minute) and the convenience of cloud upload and report by the EyeArt AI adds to the efficacy of this community-based DR screening system. EyeArt can be used by screeners to refer patients for DR when an ophthalmologist may not be readily available. Therefore, it can be deployed in non-ophthalmology medical settings, such as an endocrine clinic or emergency room, to address specific populations of patients at risk for DR.

Alongside EyeArt's high level of accuracy, it still has some limitations in older patients with cataracts. Uploading images to AI with accurate results, without the need of an onsite ophthalmologist for screening, could revolutionize the way retinal screenings are conducted, especially in times of a pandemic when patient and/or physician availability and contact may be limited. Future examination of EyeArt software could examine concordance between human graders and the EyeArt DR grades (more than mild, vision-threatening DR). Study of EyeArt deployment could also evaluate patient trust and comfort in the integration of artificial intelligence into screening services. Overall, EyeArt may play a major role in extending retinal screenings and increasing pick-up

in individuals with T2DM, especially in underserved communities.

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Declarations

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Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent to publish Not applicable.

References

1. Prevention CFDCa (2022) National diabetes statistics report. National Diabetes Statistics Report Website
2. Control CFD (2020) Prevalence of prediabetes among adults. Diabetes. CDC
3. National Diabetes Statistics Report (2020) Centers for disease control and prevention
4. Gruss SM, Nhim K, Gregg E, Bell M, Luman E, Albright A (2019) Public health approaches to type 2 diabetes prevention: the US National Diabetes Prevention Program and beyond. *Curr Diabetes Rep* 19:1–11

5. Lawrenson JG, Graham-Rowe E, Lorencatto F, Burr J, Bunce C, Francis JJ et al (2018) Interventions to increase attendance for diabetic retinopathy screening. *Cochrane Database Syst Rev* 1:CD012054
6. Cha AE, Villarroel MA, Vahratian A (2019) Eye disorders and vision loss among US adults aged 45 and over with diagnosed diabetes, 2016–2017
7. Hudson SM, Modjtahedi BS, Altman D, Jimenez JJ, Luong TQ, Fong DS (2022) Factors affecting compliance with diabetic retinopathy screening: a qualitative study comparing English and Spanish speakers. *Clin Ophthalmol* 16:1009–1018
8. Grzybowski A, Brona P, Lim G, Ruamviboonsuk P, Tan GS, Abramoff M et al (2020) Artificial intelligence for diabetic retinopathy screening: a review. *Eye* 34(3):451–460
9. Coney JM, Scott AW (2022) Racial disparities in the screening and treatment of diabetic retinopathy. *J Natl Med Assoc* 114(2):171–181
10. Fathy C, Patel S, Sternberg P Jr, Kohanim S (2016) Disparities in adherence to screening guidelines for diabetic retinopathy in the United States: a comprehensive review and guide for future directions. *Semin Ophthalmol* 31:364–377
11. Stark A (2018) FDA permits marketing of artificial intelligence-based device to detect certain diabetes-related eye problems. FDA, Silver Spring
12. Eyenuk (2020) Eyenuk announces FDA clearance for EyeArt autonomous AI system for diabetic retinopathy screening. <https://www.eyenuk.com/us-en/articles/diabetic-retinopathy/eyenuk-announces-eyear-art-fda-clearance/>. Accessed 2022
13. Bhaskaranand M, Ramachandra C, Bhat S, Cuadros J, Nittala MG, Sadda SR et al (2019) The value of automated diabetic retinopathy screening with the EyeArt system: a study of more than 100,000 consecutive encounters from people with diabetes. *Diabetes Technol Ther* 21(11):635–643
14. Thylefors B, Chylack L Jr, Konyama K, Sasaki K, Sperduto R, Taylor H et al (2002) A simplified cataract grading system the WHO cataract grading group. *Ophthalmol* 9(2):83–95
15. Lim JJ, Regillo CD, Sadda SR, Ipp E, Bhaskaranand M, Ramachandra C et al (2023) Artificial intelligence detection of diabetic retinopathy: subgroup comparison of the EyeArt system with ophthalmologists' dilated examinations. *Ophthalmol Sci* 3(1):100228
16. Ipp E, Liljenquist D, Bode B, Shah VN, Silverstein S, Regillo CD et al (2021) Pivotal evaluation of an artificial intelligence system for autonomous detection of referable and vision-threatening diabetic retinopathy. *JAMA Netw Open* 4(11):e2134254
17. Vought R, Vought V, Taylor P, Szirth B, Khouri AS. Tele-robotic ophthalmic consultation
18. Sommer AC, Blumenthal EZ (2020) Telemedicine in ophthalmology in view of the emerging COVID-19 outbreak. *Graefes Arch Clin Exp Ophthalmol* 258:2341–2352

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